

Cooper Koers

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PROFILE

Motivated student-athlete at Princeton University with a curiosity-driven passion for computational science, genetics, and systems-oriented biology. Engaged in bioinformatics research, focusing on uncovering patterns in genetic data, with work spanning evolutionary genomics to somatic mutation research. Combines expertise in quantitative genetics, functional genomics, computational tools, and machine learning.

EDUCATION

Princeton University	A.B. Ecology & Evolutionary Biology , minor in Quantitative & Computational Biology	2022-2026
Relevant Coursework:	Computational Molecular Biology, Biostatistics, Genomics, Molecular Biology, Evolutionary Biology, Multivariate Calculus	
GPA:	3.5/4.0	
Athletic Achievement:	Princeton University Varsity Football Team	
	- Contributing member of team – 3-year starter for NCAA D1 program	
	- Commit 30+ hours weekly to practices, meetings, film analysis, workouts, and games	

RESEARCH EXPERIENCE

Joshua Akey Lab at Princeton Lewis-Sigler Institute for Integrative Genomics	2023 - Present
Princeton University, Princeton, NJ	
Comparative Analysis of Data in Neanderthal and Denisovan Samples with Human DNA	Summer 2023
- Conducted research into archaic human phylogenies from DNA SNP distribution across human genome	
Simulation Analysis of Evolutionary Pressures in Somatic Mutation Data	Fall 2023 – Fall 2024
- Explored implications of Ka/Ks values for somatic mutation datasets regarding evolutionary pressures	
- Developed software for simulation data used in paper under review	
Full Non-canonical Tumor Transcriptome Analysis	Summer 2024 – Present
- Exploring trends within non-canonical transcripts from cancer databases in preparation for manuscript	
Bridgett vonHoldt Lab at Princeton Ecology and Evolutionary Biology	Spring 2025 – Present
Princeton University, Princeton, NJ	
- Conducting Senior Thesis research about selective mating in American Gray Wolf populations for extinct Red Wolf-originating introgressed DNA	

INDUSTRY EXPERIENCE

Genetic Medicine Research Intern at Eli Lilly and Company	Summer 2025
Lilly Seaport Innovation Center, Boston, MA	
- Developed in house tool to optimize cDNA sequences that improved upon previous methods using a combination of language models and algorithmic design	
- Software developed utilized in current genetic medicine early-stage drug development pipeline	
Founder of Organic Intelligence	Summer 2025 – Present
Princeton, NJ	
- Exploring polygenic disease phenotypes using multi-omics techniques to construct interactomes for individualized precision genetic medicine	

PUBLICATIONS

Huixin Xu, Rob Bierman, Dayna Akey, **Cooper Koers**, Troy Comi, Claire McWhite, Joshua M. Akey. "Landscape of human protein-coding somatic mutations across tissues and individuals." bioRxiv, <https://doi.org/10.1101/2025.01.07.631808>. Submitted January 2025.

Cooper Koers, Rob Bierman, Joshua M. Akey. *MutagenesisForge: A command line tool for determining evolutionary pressure from simulated K_a/K_s ratio analysis*. Manuscript in submission, 2025.

Cooper Koers, Cara Weisman, Rob Bierman, Joshua M. Akey. *An encompassing analysis of non-canonical transcription in cancer cells*. Manuscript in preparation, 2025.

Cooper Koers, Joshua M. Akey. *How robots may reproduce: exploring robotic evolution through the framework of Lambda Calculus*. Manuscript in submission, 2025.

PROJECTS

MutagenesisForge

Winter 2024 – Present

- A command line interface and Python package aimed at developing biologically relevant null model dN/dS statistics

LambdaForge

Spring 2025 – Present

- Python package dedicated to modelling reproduction and recombination between Lambda expressions to mimic reproduction of logical statements

SKILLS

Programming Languages: Python, R, Java, shell scripting, SLiM

Bioinformatics Tools: BLAST, GATK, SLURM workload manager, Samtools, Bedtools

Laboratory Techniques: PCR, gel electrophoresis, next-generation DNA sequencing

Data Analysis: statistical analysis, data visualization